

Types of circuit breakers

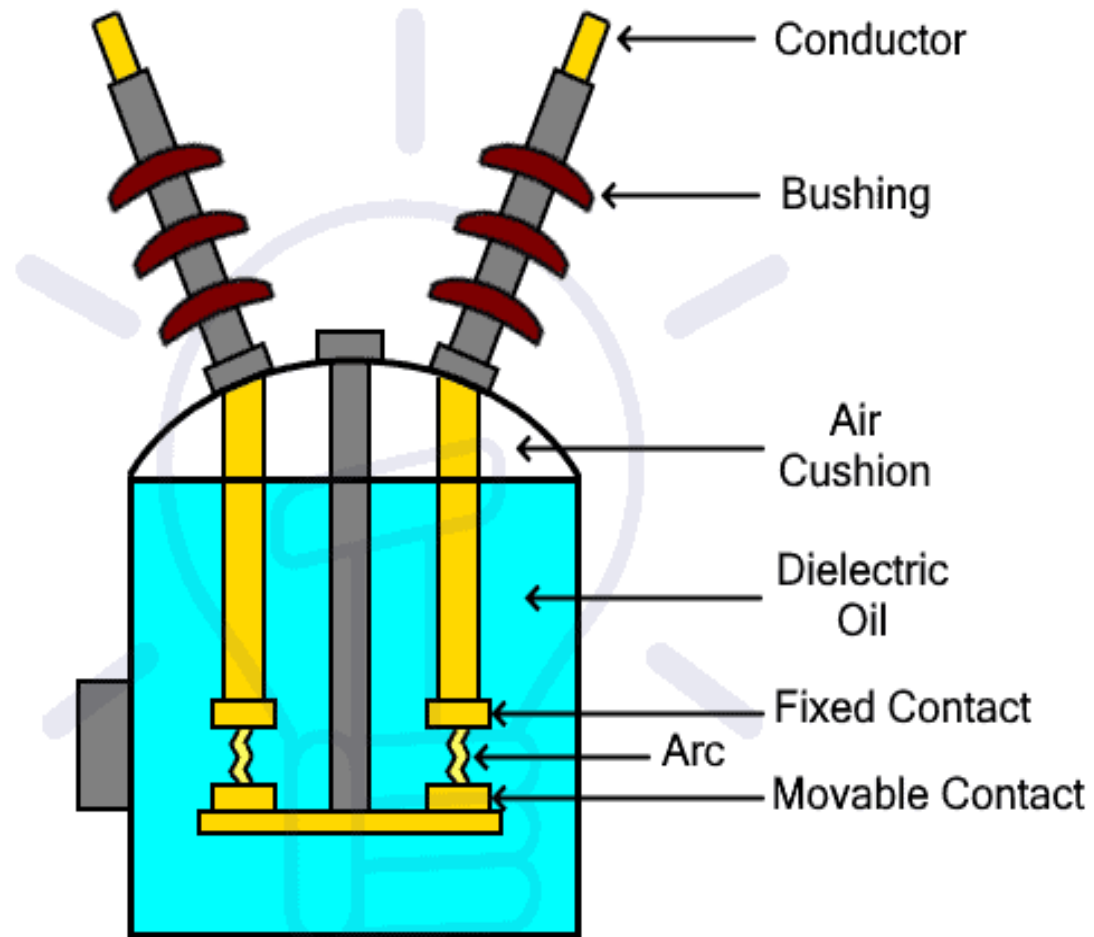
Lecture 5

High Voltage AC Circuit Breaker

- The definition of high voltage depends on context. IEC considers high voltages as the voltage that exceeds 1000v. Such voltage tends to generate an arc that is not easily extinguishable. Circuit breakers used for making and breaking contacts at such voltages are called HV circuit breakers.
- The arc extinction can be done using various methods at such high voltages. The HV circuit breaker may or may not use OIL for arc extinction; therefore, they are classified into two types:
 - 1- Oil Circuit Breaker
 - 2- Oil-Less Circuit Breaker

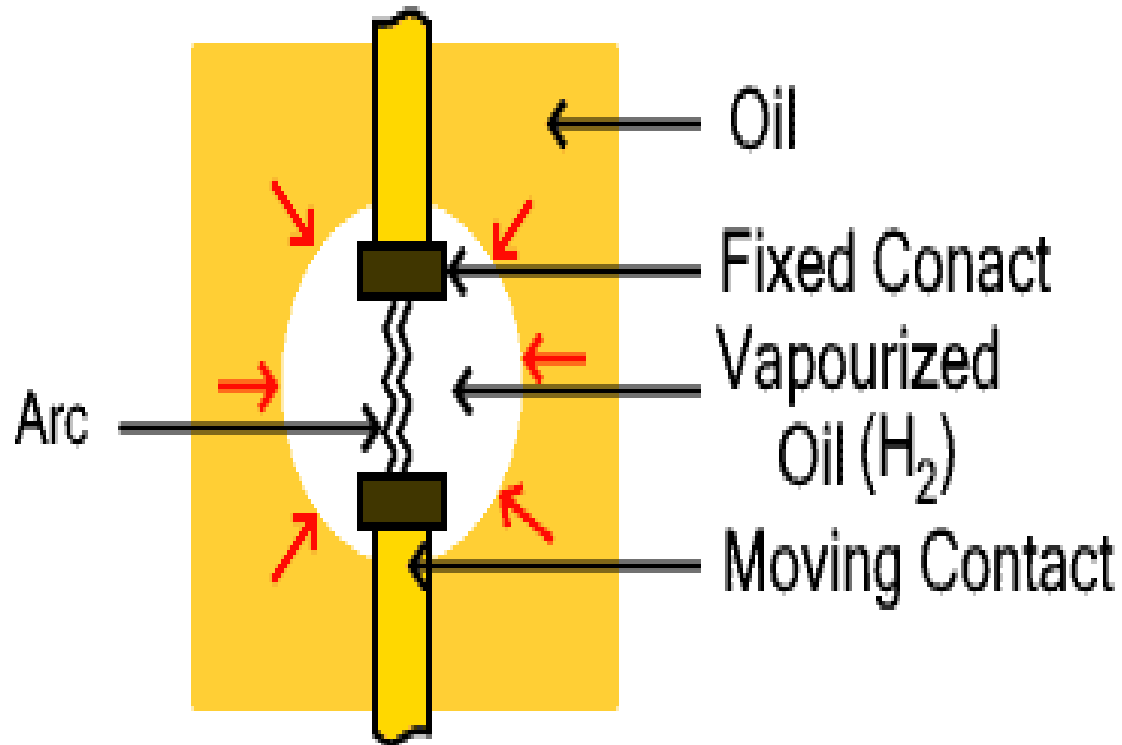
1- Oil Circuit Breaker

The type of circuit breaker that uses oil as a dielectric or insulating media to reduce the arc is called an **Oil Circuit Breaker (OCB)**. It is one of the oldest types of high voltage circuit breaker and it mainly uses the **transformer oil**. The oil used in such circuit breakers has very good insulating properties far better than air. The CB contacts are flooded in oil which is used to reduce the arc once the contacts separate. The heat generated by the arc is dissipated inside the oil.



Oil Circuit Breaker

When CB breaks its current-carrying contacts inside the oil, the distance between the contacts starts increasing. Initially, there is a very small distance between the contacts but there is also a very high voltage gradient. Due to this, the oil between the contacts starts ionizing and creates an arc between the contacts.



The arc generates a lot of heat and vaporizes the oil surrounding it which is mostly decomposed into hydrogen gas. The hydrogen gas bubbles are rapidly generated surrounding the contact almost ten times the volumes of the oil. This oil surrounding the gas bubbles put a lot of pressure on it increasing de-ionization of the media. De-ionization of the media increases its dielectric strength that will reduce the arc at zero crossing of the current. Besides that, the cooling effect of the oil and the gas bubbles also helps in arc quenching.

Based on the amount of oil being used in OCB (Oil Circuit Breaker), they are classified into two types:

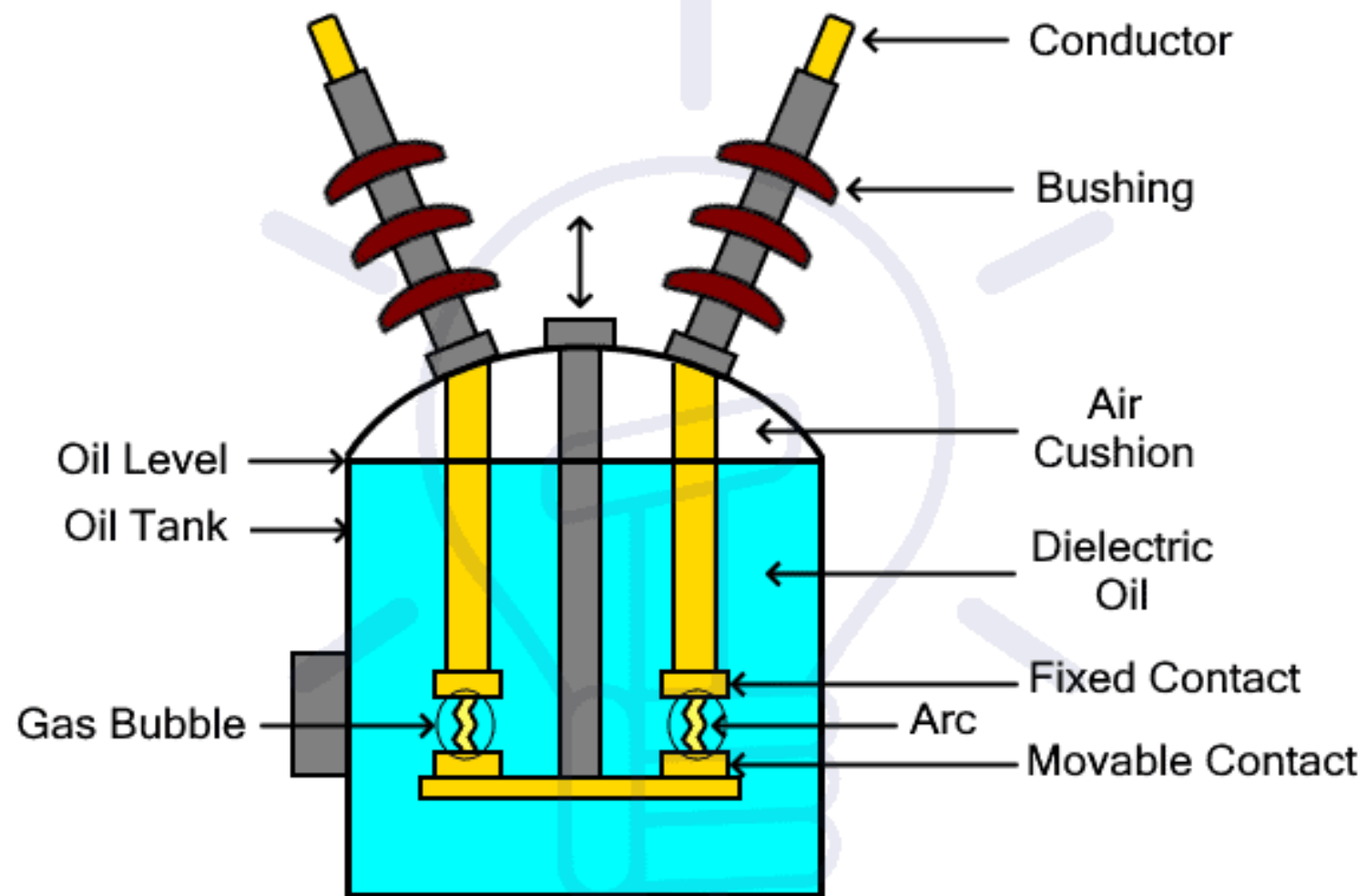
1-1- Bulk Oil Circuit Breaker (BOCB)

1-2 Minimum Oil Circuit Breaker (MOCB)

1-1- Bulk Oil Circuit Breaker (BOCB)

This type of oil circuit breaker uses the insulating oil to reduce the arc as well as insulate the live contacts from the earthed parts of the CB. Such CB uses oil in bulk.

BOCB has an iron tank that holds the insulating oil inside. The contacts (fixed and moving) are submerged in the oil. Upon breaking the contacts, the arc generates heat and produces gas. The pressurized gas displaces the oil inside the tank, where air at the top of the tank is used as a cushion. Therefore, the tank should not be completely filled with oil. Also, the tank must be strong enough to absorb the pressure released by the gas. There is also a gas vent for releasing the gas safely outside.



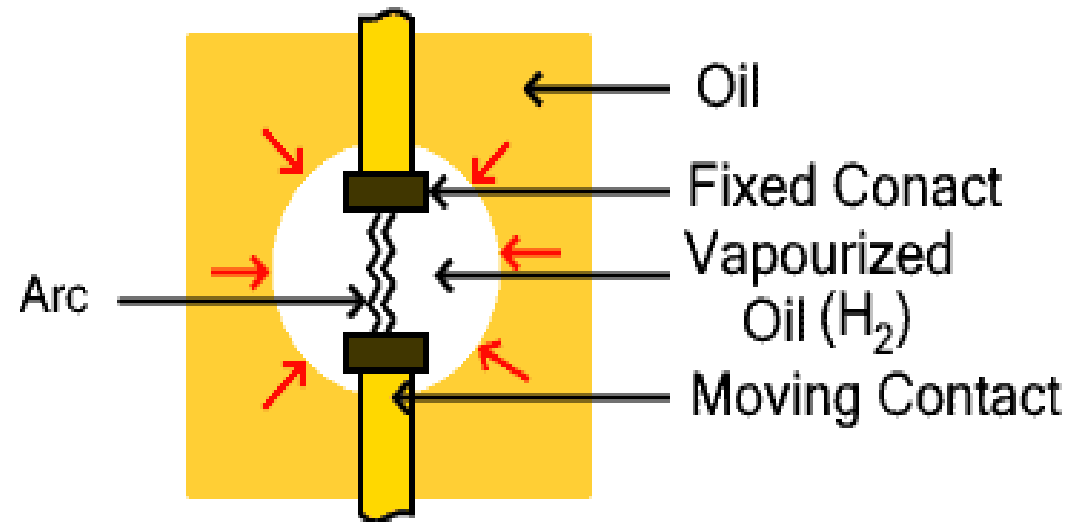
Bulk Oil Circuit Breaker

The arc is quenched by utilizing the compressed gas generated by the heat of the arc. Since the contacts are moving, the distance between the contacts is also increased. It also increases the resistance for the arc. Also, the cooling effect of the gas also plays its role in quenching the arc once the current goes through zero crossing.

The BOCB is divided into two further types **based on the division of arcs to quickly extinguish it.**

1-1-1- Single Break Bulk Oil Circuit Breaker

In single-break BOCB, **there is only one fixed contact and one moving contact.** Upon fault current, the moving contact pushes backward, generating an arc that is extinguished by the compressed gas inside the oil. As the name suggests, **there is only one break between the contacts.**



1-1-2- Double Break Bulk Oil Circuit Breaker

In double break BOCB, there are two fixed contacts and one moving contact. The fixed contacts are fixed to the tank at both ends, connected to the live conductors, while the moving contact can move upward and downward using an insulating rod.

In normal conditions, the moving contact is pushed upward to make contact at both ends with the fixed contacts. Upon fault conditions, the moving contacts are pulled downward to break the contacts and generate arcs at both ends. This way, the arc is split into two parts, having lower strength, which can be easily cooled off and reduced inside the oil.

Advantages

- The oil used for arc quenching has very high dielectric strength.
- The oil insulates the live contacts from the earthed parts.
- The oil produces hydrogen gas with the heat from the arc which is beneficial for arc quenching.
- The pressure of oil compresses the gas to deionize the media.
- The gas also helps in cooling the medium.

Disadvantages

- The oil is inflammable and causes a fire hazard.
- The contacts can get damaged with the arc.
- The carbonization of the oil from the contacts reduces its dielectric strength.
- The contacts and the oil must be checked and maintained regularly.
- The use of huge amount of oil increases its cost.
- Their large oil-filled tanks are heavy and take large space.

1-2- Minimum Oil Circuit Breaker (MOCB)

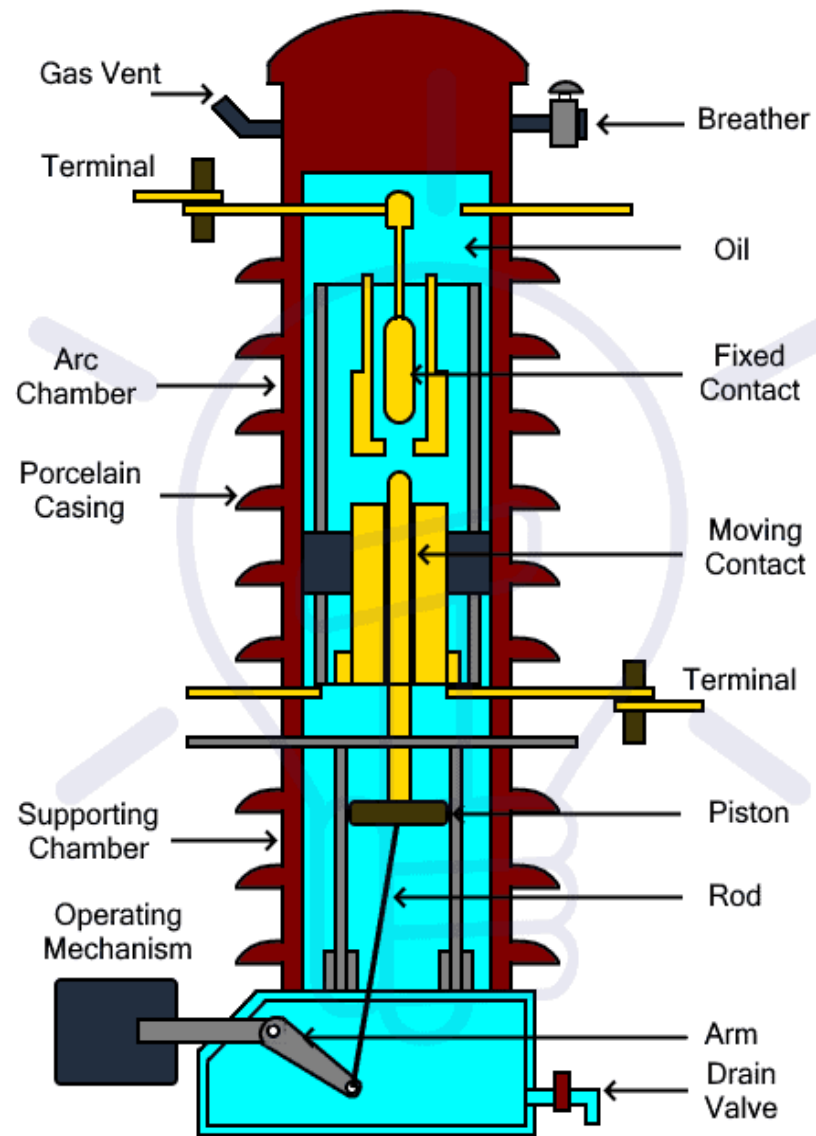
The bulk oil circuit breaker uses a huge amount of oil to quench the arc which may pose a threat for fire hazard. To reduce such risk, the MOCB uses far less oil than in BOCB. The oil is only used for arc quenching and not to isolate the live parts from earthed parts.

MOCB has two chambers: the arc chamber and the supporting chamber. The arc chamber is made from porcelain encased with backlisted paper. It is filled with insulating oil. This chamber is used for quenching the arc. It contains fixed and moving contacts.

The supporting chamber is made from porcelain mounted on top of a metal chamber. This chamber is used for isolating the arc chamber as well as support the arc chamber by mounting it on top of it. this chamber is also filled with oil used only for insulation.

The moving contact moves upwards and downwards with the help of a fixed arm in the supporting chamber. The moving contact has a fixed piston which is used for forcing the oil upwards helping to quench the arc.

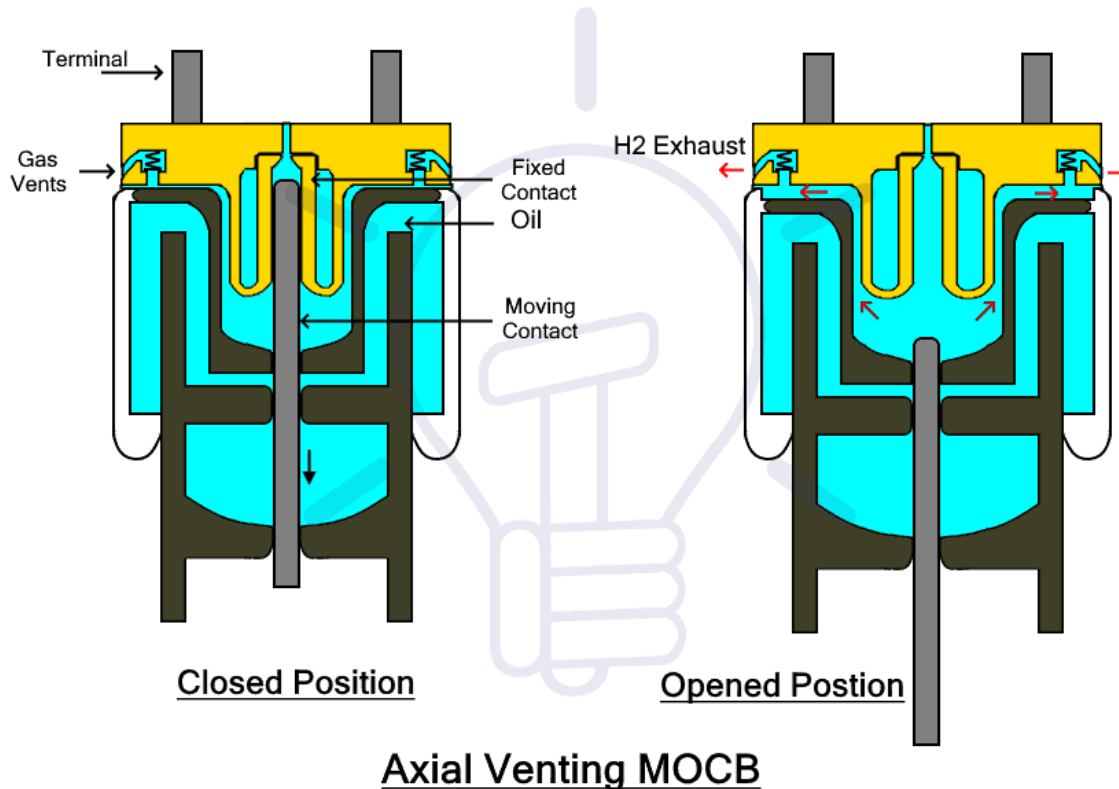
Under normal conditions, the lower moving contact makes a connection with the upper fixed contact. Upon fault condition, the arm pulls the moving contact downward and an arc is generated. This arc is extinguished by the pressurized gas in the oil surrounding it and by pushing the oil from the support chamber with the help of the piston. As the contact moves downward, a vent becomes available for the hydrogen gas to exhaust out.



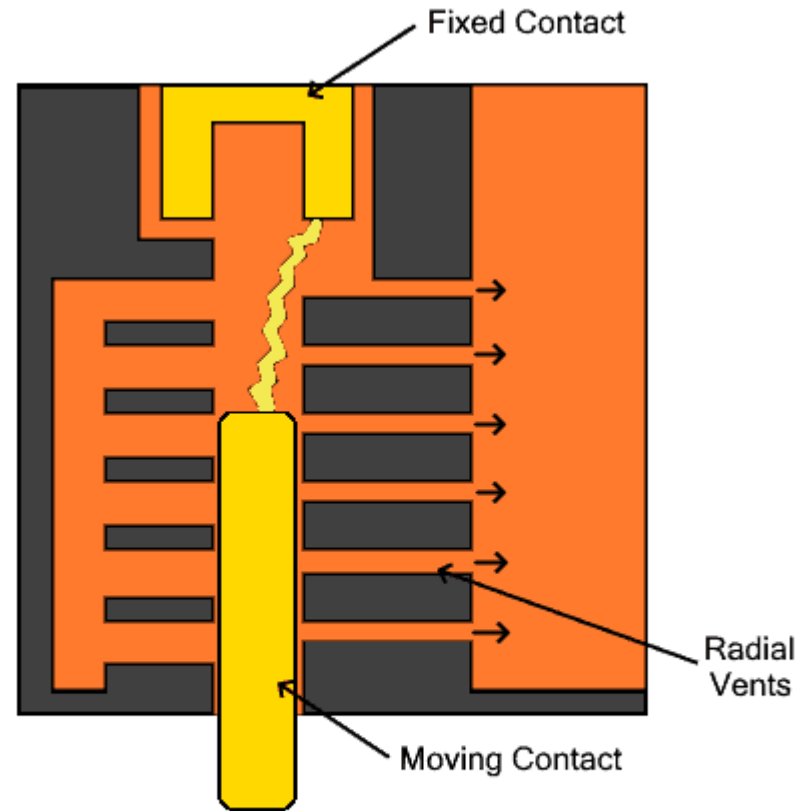
Minimum Oil Circuit Breaker

In terms of venting, MOCB is divided into two types.

1-2-1- Axial Venting MOCB: In axial venting MOCB, the arc chamber vents are designed in such a way that when the moving contacts slide downward to break the circuit, it allows an opening for the cool oil to move in from the lower vent, which sweeps the arc in the axial direction through the upper vent.



1-2-2 Radial Venting MOCB: While in radial venting MOCB, the vents are designed outward along the arc chamber. Therefore, the arc is swept outward through the multiple vents.



Radial Venting MOCB

The radial venting has low pressure; thus, the oil has low dielectric strength. Therefore, radial venting MOCB is used for high current at low voltages. A combination of both axial and radial venting is used inside a single MOCB to break both low and high voltages efficiently.

Advantages

- It requires a very small amount of oil.
- A smaller amount of oil means the risk of fire hazard is low.
- It has low weight.
- It has a small size and takes up a small space.
- It is cheaper than BOCB.
- It has easier maintenance, and oil can be easily replaced.
- Best for installation in places where it is not frequently used.

Disadvantages

- . Lesser oil is more affected by the amount of carbonization from the contacts.
- . The oil quickly loses its dielectric strength.
- . It requires more frequent maintenance.